

PRACTICAL STRATEGIES

Neural Networks

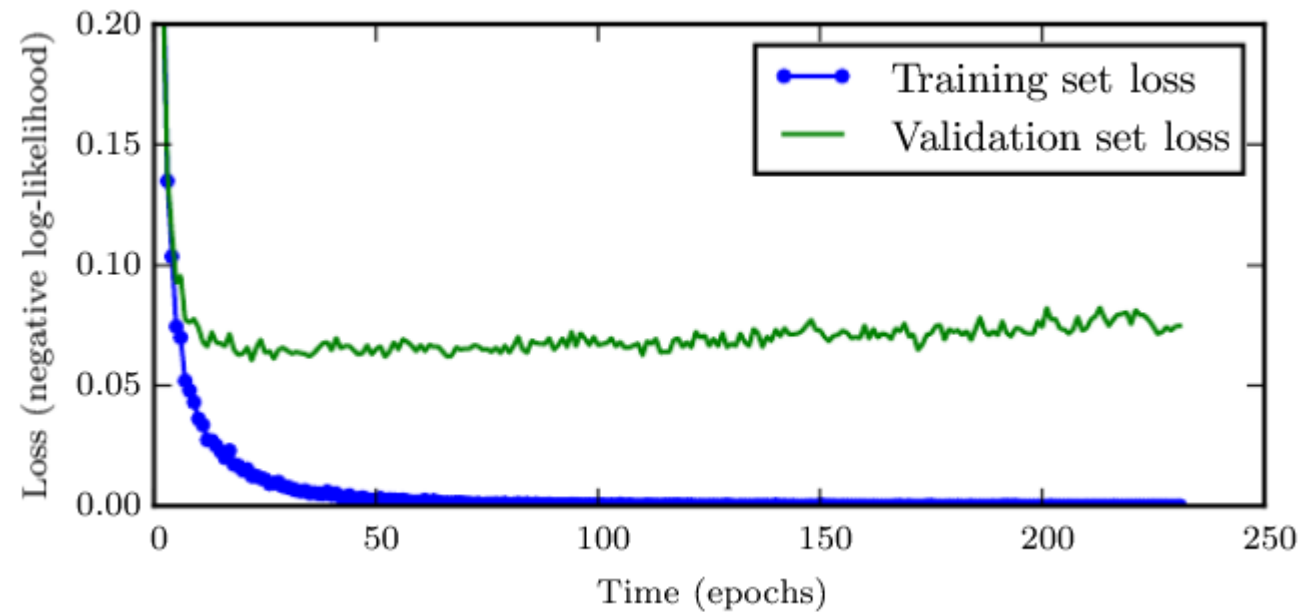
WHAT IS THE ISSUE?

		Validation set performance	
		Bad	Good
Training set performance	Bad		
	Good		

HOW DO WE KNOW WHAT IS GOOD/BAD?

- Compare with previous/baseline models
- Compare with human performance
- Practical constraints from application

LEARNING CURVES



Goodfellow et al., Deep Learning Book, 2016

WHAT IS THE ISSUE?

		Validation set performance	
		Bad	Good
Training set performance	Bad	Underfitting	
	Good		

WHAT TO DO ABOUT UNDERFITTING?

- Increase model capacity
- Train model longer
- Tune learning rate
- Pre-processing

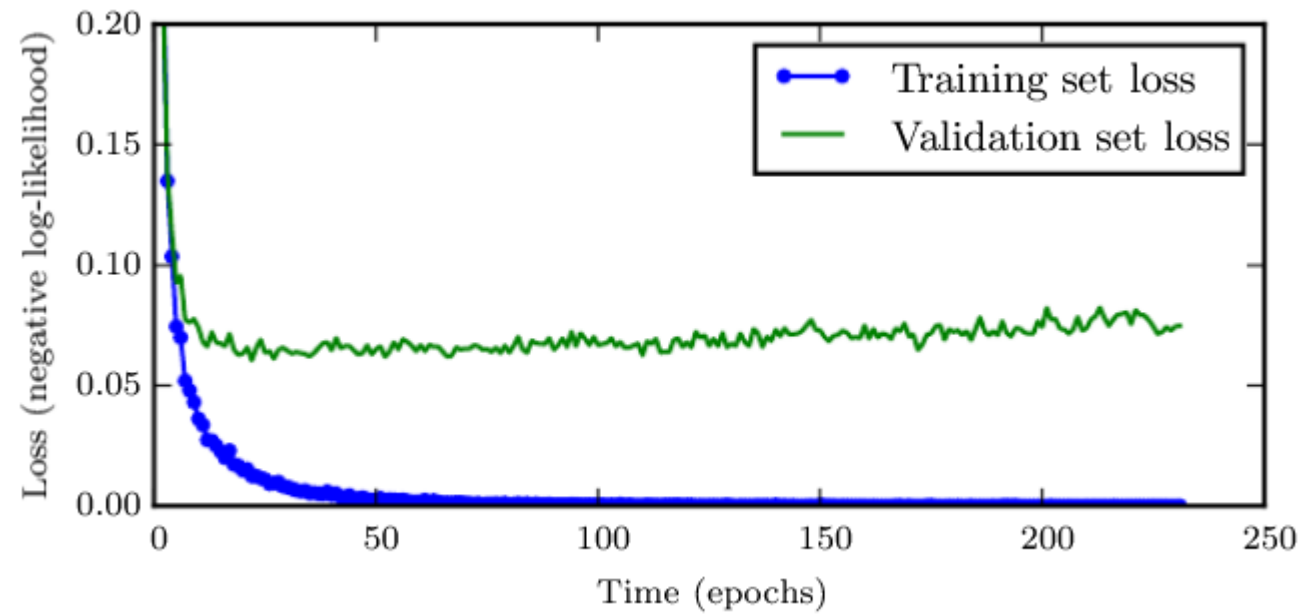
MODEL CAPACITY



Hyperparameter	Increases capacity when...	Reason	Caveats
Number of hidden units	increased	Increasing the number of hidden units increases the representational capacity of the model.	Increasing the number of hidden units increases both the time and memory cost of essentially every operation on the model.
Learning rate	tuned optimally	An improper learning rate, whether too high or too low, results in a model with low effective capacity due to optimization failure.	
Convolution kernel width	increased	Increasing the kernel width increases the number of parameters in the model.	A wider kernel results in a narrower output dimension, reducing model capacity unless you use implicit zero padding to reduce this effect. Wider kernels require more memory for parameter storage and increase runtime, but a narrower output reduces memory cost.
Implicit zero padding	increased	Adding implicit zeros before convolution keeps the representation size large.	Increases time and memory cost of most operations.
Weight decay coefficient	decreased	Decreasing the weight decay coefficient frees the model parameters to become larger.	
Dropout rate	decreased	Dropping units less often gives the units more opportunities to “conspire” with each other to fit the training set.	

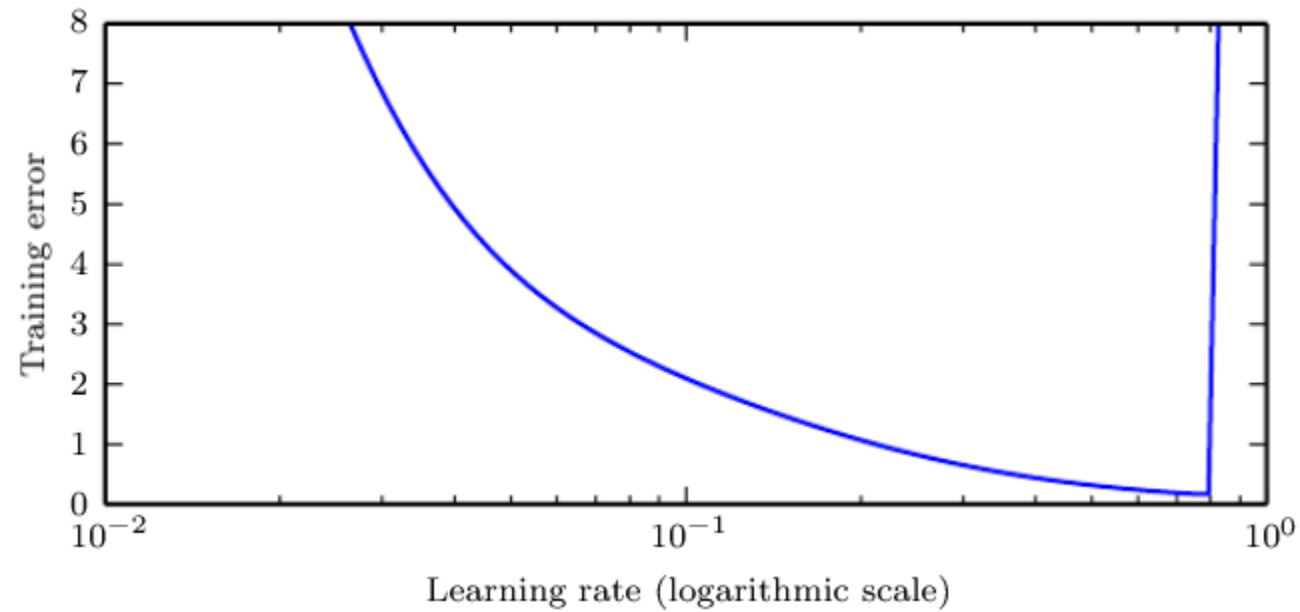
Goodfellow et al., Deep Learning Book, 2016

LEARNING CURVES



Goodfellow et al., Deep Learning Book, 2016

LEARNING RATE



Goodfellow et al., Deep Learning Book, 2016

PRE-PROCESSING



Carleton
UNIVERSITY

ADMITTING DEFEAT

- Data is too noisy
- Feature set insufficient

WHAT IS THE ISSUE?

		Validation set performance	
		Bad	Good
Training set performance	Bad		Impossible?
	Good		

WHAT IS THE ISSUE?

		Validation set performance	
		Bad	Good
Training set performance	Bad		
	Good	Overfitting	

WHAT TO DO ABOUT OVERFITTING?

- Decrease model capacity
- Early Stopping
- Regularization
- Dropout
- Batch normalization
- Collect more data

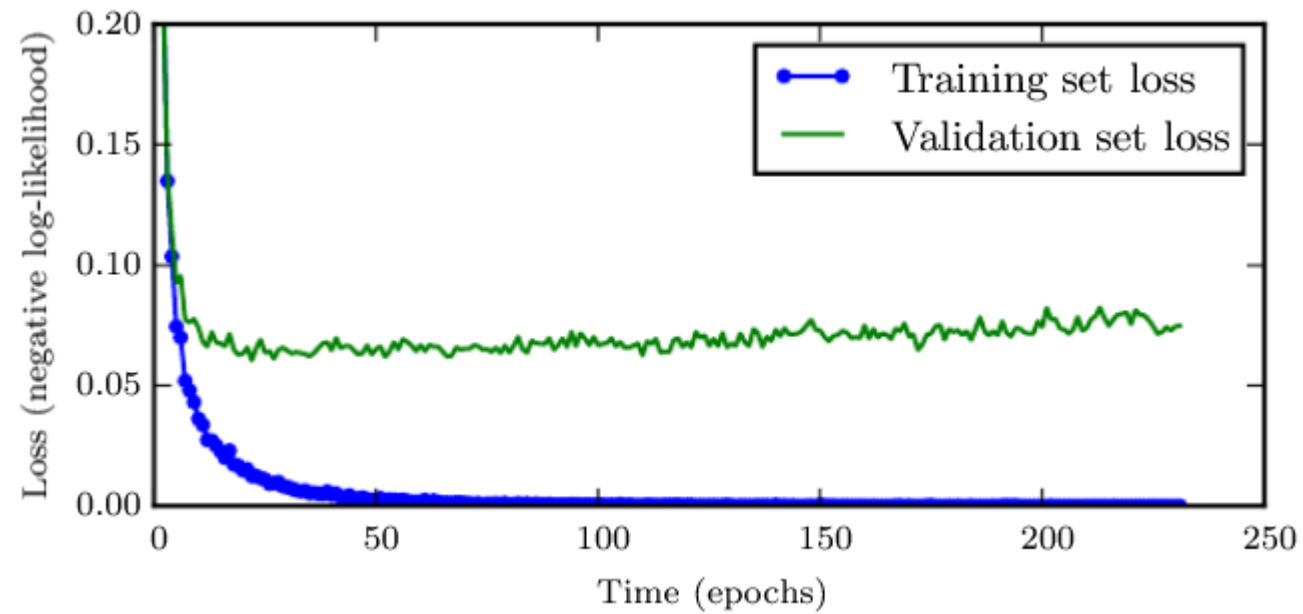
MODEL CAPACITY



Hyperparameter	Increases capacity when...	Reason	Caveats
Number of hidden units	increased	Increasing the number of hidden units increases the representational capacity of the model.	Increasing the number of hidden units increases both the time and memory cost of essentially every operation on the model.
Learning rate	tuned optimally	An improper learning rate, whether too high or too low, results in a model with low effective capacity due to optimization failure.	
Convolution kernel width	increased	Increasing the kernel width increases the number of parameters in the model.	A wider kernel results in a narrower output dimension, reducing model capacity unless you use implicit zero padding to reduce this effect. Wider kernels require more memory for parameter storage and increase runtime, but a narrower output reduces memory cost.
Implicit zero padding	increased	Adding implicit zeros before convolution keeps the representation size large.	Increases time and memory cost of most operations.
Weight decay coefficient	decreased	Decreasing the weight decay coefficient frees the model parameters to become larger.	
Dropout rate	decreased	Dropping units less often gives the units more opportunities to “conspire” with each other to fit the training set.	

Goodfellow et al., Deep Learning Book, 2016

EARLY STOPPING



Goodfellow et al., Deep Learning Book, 2016

REGULARIZATION

DROPOUT

BATCH NORMALIZATION

COLLECT MORE DATA

WHAT IS THE ISSUE?

		Validation set performance	
		Bad	Good
Training set performance	Bad		
	Good		Success!